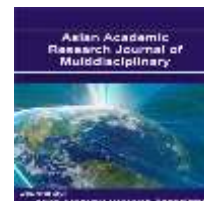




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**MAPPING OF RIVER SAND MINING ZONES USING REMOTE SENSING AND
GIS: A CASE STUDY IN PARTS OF PAPAGNI AND PENNAR RIVER BEDS, YSR
DISTRICT ANDHRA PRADESH**

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Abstract

Sand is an important engineering material for construction works. Sand deposition in river is the biggest source of sand, in recent times there has been immense pressure on Rivers which acts as a treasure for the building sector due to rapid growth in the development activities. The main objective of the study is to investigate, recognize the sand mining areas of Papagni and Pennar River in parts Kamalapuram, Kahjipeta, Vallur, Chennur and Kadapa Revenue Mandals of YSR District, Andhra Pradesh, by using Google earth temporal data for environmental monitoring and sustainable development. A total of 9 major sites were identified demarcated from Google earth and the areal extent calculated using arcGIS10 software. The largest part of the sand mining was identified at locations of Kamalapuram and Vallur Mandals, might be due to well connecting roads nearer to the sites when compared to the other sites whereas locations of Khajipet and Kadapa mandals were least affected might be due to Geographic setting(hilly region). This study is an attempt to know the utility of Google earth, Remote Sensing and GIS software tools in the field of natural resource monitoring and environmental protection.

Key Words: River sand mining, Papagni and Pennar River beds, Remote Sensing and GIS.

References

1. AliyuBaba Nabegu. (2013) The effect of sand mining on groundwater in Kano river catchment, Journal of environment and earth science Vol.3, No2. pp. 81-87.
2. Bull, W. B. and Scott, K. M. (1974) Impact of mining gravel from urban stream beds in the southwestern United States. Geology Vol. 4. pp. 171-174.
3. CGWB, (2013). Groundwater Brouchure, YSR DIstrict(Kadapa), Andhrapradesh Minisrty of Water Resources, Govt. of India.
4. ChandrasekharaRao, M. (2006), Groundwater depletion in Papagani catchment, Economic and political weekly February18.
<http://www.indiaenvironmentportal.org.in/category/17898/author/m-chandrasekhara-rao/>
5. Chrisman. N. R., Cowen, D. J., Fisher, P. F., Goodchild, M. F. and Mark, D. M. (1989) Geographic Information Systems. In Geography in America, ed. G. L. Gaile and C. J. Willmott, pp. 776-796. Merrill, Columbus. OH.
6. Google earth : <https://earth.google.com/>
7. Fuller, I. C., Large, A. R. G. and Milan, D. J. 2003 “Quantifying Channel Development and Sediment Transfer Following Chute-Off in a Wandering Gravel-Bed River,” Geomorphology, Vol. 54, No. 3-4, , pp. 307-323.
8. Hemalatha, A.C., Chandrakanth, M.G. and Nagaraj, N. (2005): Effects of sand mining on groundwater depletion in Karnataka.
<http://ageconsearch.umn.edu/bitstream/43619/2/sand%20mining%20hemalatha%2c%20mgc%2c%20nn%20article.pdf>. Accessed on June 20, 2013.
9. Kondolf, G. M. (1994) Geomorphic and environmental effects of instream gravel mining. Landscape and Urban Planning Vol. 28, 225-243.
10. Kumm, M., Lub, X. X., Rasphonec, A., Sarkkulad , J. and Koponen, J. (2007) “Riverbank Changes along the Mekong River: Remote Sensing Detection in the Vientiane-Nong Khai Area,” Quaternary International, Vol. 186, No. 1, 2008, pp. 100-112.
11. Lee. H. Y., Deng-Tsuang, F. and Song. M. H. (1993) Migration of rectangular mining pit composed of uniform sediment. Journal of Hydraulic Engineering Vol. 119(1), 64-80.
12. Leeuw, J.D., Shankmam, D., Wu, G., De Boer, W.F., Burnham, J., HeQing, Yesoy, H. and Xiao, J. (2010): Strategic assessment of the magnitude and impacts of sand mining in Poyang lake, China. Regional Environmental Change, Vol. 10: 95-102.
13. Li, L. Q., Lu, X. X and. Chen, Z (2007) River Channel Change during the Last 50 Years in the Middle Yangtze River: An Example of the Jianli Reach, Geomorphology, Vol. 85, No. 3-4, , pp. 185-196.

14. Marh, B.S and Pathania, S.S. (2008): Environment degradation due to slate mining: a case study of Khanyara Mines, Kangra District, Himachal Pradesh. In Perspectives in Resources Management in Developing Countries, Vol. 2, Part 1: Population, Resources and Development), ed., Thakur, B., Concept Publishing Company, New Delhi, India: 232-243.
15. Mayer. P. G. (1972) The hydraulic and hydrologic effects of commercial in-stream dredging in the Allegheny River. Statement prepared for the Upper Allegheny River Dredging Group and presented before the hearing panel of the Environmental Resources Board, Commonwealth of Pennsylvania, Franklin. PA.pp. 28-29.
16. Padmalal, D., Maya, K., Sreebha, S. and Sreeja, R. (2008): Environmental effects of river sand mining: a case from the river catchment of Vembanad Lake, Southwest coast of India. Environmental Geology, Vol. 54: 879-889.
17. Raper, I. (1991) Geographical information systems. Progress in Physical Geography Vol. 15(4), 438-444.
18. Rinaldi, M.(2003,) Recent Channel Adjustments in Alluvial Rivers of Tuscany, Central Italy, Earth Surface Processes and Landforms, Vol. 28, No. 6, pp. 587-608.
19. Singh, G., Mehta, K.K., Sharma, R.C., Chawla, K.L., Joshi, P.K. and Yaduvanshi, N.P.S. (2007) Sand mining or no mining in agriculture fields in Haryana. Central Soil Salinity Research Institute, Karnal, Haryana, India, Technical Report: 1-28.
20. Subrata Pan (2013) Application of Remote Sensing and GIS in studying changing river course in Bankura District, West Bengal. International Journal of Geomatics and Geosciences Vol.4 (1), pp.149-163.
21. [Sudhir](#), S. N. V. (2016) Andhra Pradesh pushes crushed rock as sand, DECCAN CHRONICLE. Feb 17, 2016.)
22. Surian, N.(1999) Channel Changes Due to River Regulation: The Case of the Piave River, Italy, Earth Surface Processes and Landforms, Vol. 24, No. 12, pp. 1135- 1151.
23. Tepordei, V.V. (1995): National demand for crushed stone and sand and gravel. West Virginia Geological and Economic Survey, Vol. 46: pp.226.
24. Woodward-Clyde Consultants (1980)Gravel Removal Studies in Arctic and Subarctic Floodplains in Alaska. Technical Report, US Fish and Wildlife Service, Washington, DC
25. Yang, X., Damen, M. C. J. and van Zuidam, R. A. (1999,) Satellite Remote Sensing and GIS for the Analysis of Channel Migration Changes in the Active Yellow River Delta, China, International Journal of Applied Earth Observation and Geoinformation, Vol. 1, (2), pp. 146-157.